Predictions of Small-Body Physical Traits Based on Orbital Characteristics

MICRO PROJECT 1

Small Body Project Repo

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October 11, 2026

AI Use Disclosure & Academic Integrity Compliance

In compliance with the **National University Student Guidelines for Generative AI**, this project discloses the following tool usage:

- **Brainstorming & Dataset Selection (No Citation Required): Google Gemini** and **OpenAI ChatGPT** were utilized to brainstorm and isolate a viable machine learning regression scope project for the author selected JPL NASA Small-Body Dataset.
- **Data Selection & Filtering (Informed by AI): Google Gemini** acted as a technical guide to filter query parameters on the JPL NASA Small-Body Dataset web interface. Based on these logical constraints, a custom CSV file was generated and downloaded manually, which is preserved in the project's GitHub repository.
- **Verbiage & Phrasing Refinement (Informed by AI): Google Gemini** and **OpenAI ChatGPT** were utilized as writing assistants to polish phrasing and improve technical sentence flow within the title, subtitle, description, problem statement, hypothesis formulation, acquire, and prepare sections. All underlying intellectual ideas and scientific goals remain entirely of the author.
- **Modified Text & Assisted Paragraphs (Cited Inline):** Select technical paragraphs where AI-generated text was slightly modified or adapted by the author are explicitly attributed with inline citations `(OpenAI,`

`2026)` or `(Google, 2026)` directly within the text documentation.

- **Creative Material (Direct Quotes and AI-Generated & Author Modified):** Introductory "flavor text" and background narrative elements are either direct quotes by **OpenAI ChatGPT** and **Google Gemini** or generated by **OpenAI ChatGPT** and **Google Gemini** with modifications by the author. These sections are explicitly marked with `[AI DIRECT QUOTE - FLAVOR TEXT]` or `[AI GENERATED & AUTHOR MODIFIED - FLAVOR TEXT]` headers, are included strictly for entertainment purposes only, and are excluded from the graded scientific material.
- **Image Generation (AI-Generated): Midjourney** was utilized to create conceptual and illustrative images for the project. These AI-generated images are used strictly for visual enhancement and aesthetic purposes, and are not presented as actual astronomical data, original scientific observations, or graded scientific material.
- **Disclosure Formatting & Meta-Assistance (AI-Generated & Author Modified): Google Gemini** was utilized to structure, edit, and generate the text of this disclosure block to ensure compliance with institutional requirements. This entire block is AI-generated and modified by the author.

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1. Problem Statement

State the occurrence that you are attempting describe using data science

> **[AI GENERATED & AUTHOR MODIFIED - FLAVOR TEXT]**
> Physical prospecting and exploration of small-bodies incur heavy operational and capital costs, making the early detection of non-viable targets critical. Small-body volumetric size is a primary variable for predicting economic viability. A small-body that fails to meet the minimum size threshold will not meet viability requirements even if the

> small-body has exceptional mineral composition. To reduce the risk of prospecting and exploring non-viable small-bodies, a data model must be created to filter out sub-threshold small-bodies.

Peer-reviewed research has determined certain orbital characteristics such as albedo, absolute magnitude *(H)*, semi-major axis *(a)*, eccentricity *(e)*, and inclination *(i)* can be used to increase accuracy for estimating small-body size (Ge, 2026). These characteristics have been broken down into three broad categories based on the relative cost and reliability of observing these characteristics. Group 1 characteristics are low cost to observe and the observations have high reliability (Murcia, 2023). Group 2 characteristics are low cost to observe but have medium reliability because the observations are highly variable and prone to observational error when gathering (Robinson, 2024). Group 3 characteristics are high cost to observe and have high reliability (Ge, 2025).

The model must be cost-effective in determining small-body size accurately. Group 3 characteristics are unlikely to be cost-effective and there is limited access to the equipment because access is allocated based on competing use requests from other higher priority projects (NASA, 2025). Group 1 and Group 2 characteristics provide the best candidates for cost-effective observations, however, since Group 2 includes characteristics that are prone to observational error a cost-benefit analysis will determine if excluding these observations will decrease accuracy in determining the size of a small-body beyond the statistical and operational acceptable thresholds.

A two-model comparison is necessary to determine what model will create the most cost-effective system for filtering out sub-threshold small-bodies. The first model will use regression analysis via machine learning with Group 1 and Group 2 characteristics to predict the diameter of a small-body. The second model will act as a scientific baseline, using regression analysis via machine learning with Group 1 characteristics only to predict the diameter of a small-body. Because current peer-reviewed research suggests absolute magnitude *(H)* (Group 2 characteristic) is mathematically required to calculate diameter (Ge, 2026), this analysis will determine if relying solely on highly reliable Group 1 characteristics results in a failure to accurately predict diameter. Each model will be assessed for overall accuracy to predict the size of a small body. An analysis of the accuracy to predict the diameter of small-body will determine if the operational trade-off between feature measurement cost and predictive performance of using Group 2 characteristics will have a material impact on properly filtering sub-threshold small-bodies.

Once the two-model comparison is complete the results will be used to create the final model using the more cost-effective characteristics within the acceptable accuracy range.

2. Hypothesis Formulation

Suggested explanation to the problem statement.

The project utilizes a two-model comparison framework that will be assessed from a statistical and operational perspective. The regression models target diameter *(Y)* using two feature subsets: an expanded set of low-cost, high reliability observations with low-cost, medium reliability observations ($\mathbf{X}_{full}$) and a restricted set of only low-cost, high reliability observations ($\mathbf{X}_{restricted}$). While the restricted model ($\mathbf{X}_{restricted}$) is preferred due to the lower number of features and high reliability, current peer-reviewed research dictates that absolute magnitude *(H)* from Group 2 is mathematically required to calculate diameter (Ge, 2026). As such, the framework will determine if dropping the less reliable Group 2 characteristics results in statistically and operationally significant degradation in accuracy to predict target diameter which then justifies the observational penalties of the expanded model ($\mathbf{X}_{full}$).

The hypothesis determines if dropping the less reliable Group 2 characteristics falls within an acceptable operational threshold (δ), quantified as:

$$\Delta R^2_{test} = R^2_{Model(\mathbf{X}_{full})} - R^2_{Model(\mathbf{X}_{restricted})}$$

Statistical Hypothesis

The statistical hypothesis tests if the restrictive feature subset of Group 1 characteristics maintains a statistically acceptable degree of accuracy in determining target diameter *(Y)* relative to the expanded feature subset of Group 1 and Group 2 characteristics.

- **Statistical Null Hypothesis ($H_{0,stat}$):** $\Delta R^2_{test} \leq 0$

  The inclusion of Group 2 features results in no improvement or a decrease in the out-of-sample ΔR^2 on the testing dataset over the baseline model on Group 1 features only.
- **Statistical Alternative Hypothesis ($H_{1,stat}$):** $\Delta R^2_{test} > 0$

  The inclusion of Group 2 features yields a statistically significant increase in the out-of-sample ΔR^2 on the testing dataset, indicating that the expanded features materially increase the accuracy of predicting the target diameter *(Y)*.
- **Evaluation Criteria:** Non-parametric permutation testing or bootstrapping procedures will be utilized in evaluation to calculate the empirical p-value or confidence intervals for ΔR^2_{test} to validate statistical significance.

Operational Hypothesis

The operational hypothesis tests if the restrictive feature subset of Group 1 characteristics maintains an operationally practical acceptable degree of accuracy (δ) in determining target diameter *(Y)* relative to the expanded feature subset of Group 1 and Group 2 characteristics.

- **Operational Null Hypothesis ($H_{0,ops}$):** $\Delta R^2_{test} \leq \delta$
  The performance gain from adding Group 2 features is less than or equal to the operational threshold (δ). The marginal accuracy increase does not justify the added complexity and increase in number of features relative to the baseline model of Group 1 features only.
- **Operational Alternative Hypothesis ($H_{1,ops}$):** $\Delta R^2_{test} > \delta$
  The performance gain from adding Group 2 features exceeds the operational threshold (δ). The accuracy improvement materially outweighs the tracking overhead and penalty making the expanded model ($\mathbf{X}_{full}$) optimal for production.

3. Acquire

Identify data sets, retrieve data, query data.
In order to test the two-model comparison framework, a comprehensive and reliable dataset is required. The dataset needed to contain characteristics from Group 1 and Group 2 for small-bodies. The JPL NASA Small-Body Dataset was selected because NASA is an authoritative, reliable, and primary repository of small-body data that is also open-source and available to the public allowing reproduction of testing for peer-review.

The JPL NASA Small-Body Dataset contains Group 1 and Group 2 characteristics such as absolute magnitude *(H)*, semi-major axis *(a)*, eccentricity *(e)*, and inclination *(i)* which directly relate to the problem statement and hypothesis requirements necessary for the expanded features subset and restricted feature subset.

The JPL NASA Small-Body Dataset was queried to meet the target variable diameter *(Y)* to be used in out-of-sample testing. The query exclusively included asteroids with a known diameter greater than zero kilometers (0 km). This ensures a valid target variable for regression training and the calculation of the coefficient of determination *(R^2)*. Based on these parameters, a custom CSV file was generated and downloaded manually, which is preserved in the project's GitHub repository.

4. Prepare

Explore and pre-process.
Data has been prepared for regression analysis in the following ways:

1. **Target Variable Isolation (diameter):** All entries without a diameter have been removed. Diameter *(Y)* is the target variable and is required for the training model and to calculate the coefficient of determination *(R^2)* during out-of-sample testing.

2. **Exclusion of High-Cost Features (albedo):** Albedo represents a Group 3 characteristic and was explicitly dropped from the dataset. Removing this feature ensures the model does not rely on cost-prohibited data and maintains operational cost-effective analysis. This also prevents data leakage between albedo and the target diameter.
3. **Feature Subset Selection:** The dataset was restricted to specific characteristics required to construct the two-model comparative framework. This isolates the low-cost, high-reliability Group 1 characteristics *(e.g., semi-major axis a, eccentricity e, inclination i)* and the low-cost, medium-reliability Group 2 characteristic *(absolute magnitude, H)* needed to test the operational hypothesis.
4. **Feature Completeness:** Entries with missing values in the selected feature subsets were removed. This ensures the regression algorithms receive a complete feature matrix and that any observed variance in R^2_{test} is attributable to specific feature subsets and not fluctuations in sample size or missing data.

In [1]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

dataset_path = r"data/dataset-jpl-small-body-asteroids.csv"
dataset_cleaned_path = r"data/cleaned-dataset-jpl-small-body-asteroids.csv"
target_columns = ['diameter']
dropped_columns = ['albedo']
selected_columns = ['H', 'diameter', 'a', 'e', 'i', 'om', 'q', 'ad', 'per_y']

df = pd.read_csv(dataset_path)
df.head()
```

Out[1]:

	full_name	H	diameter	albedo	a	e	i	om	w	q	ad	per_y	data_arc	condition_code	n_obs_u
0	1 Ceres (A801 AA)	3.34	939.400	0.0900	2.766	0.0797	10.59	80.25	73.29	2.545	2.99	4.60	9520.0	0	
1	2 Pallas (A802 FA)	4.12	513.000	0.1550	2.770	0.2307	34.93	172.89	310.97	2.131	3.41	4.61	80828.0	0	
2	3 Juno (A804 RA)	5.19	246.596	0.2140	2.671	0.2557	12.99	169.81	247.90	1.988	3.35	4.37	80897.0	0	
3	4 Vesta (A807 FA)	3.25	522.770	0.4228	2.361	0.0902	7.14	103.70	151.47	2.148	2.57	3.63	25743.0	0	
4	5 Astraea (A845 XA)	6.95	106.699	0.2740	2.577	0.1876	5.36	141.45	359.38	2.093	3.06	4.14	65687.0	0	

In [2]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 139582 entries, 0 to 139581
Data columns (total 17 columns):
 #   Column          Non-Null Count   Dtype  
---  ------          --------------   -----  
 0   full_name       139582 non-null  str    
 1   H               139203 non-null  float64
 2   diameter        139582 non-null  float64
 3   albedo          138459 non-null  float64
 4   a               139582 non-null  float64
 5   e               139582 non-null  float64
 6   i               139582 non-null  float64
 7   om              139582 non-null  float64
 8   w               139582 non-null  float64
 9   q               139582 non-null  float64
 10  ad              139582 non-null  float64
 11  per_y           139582 non-null  float64
 12  data_arc        139577 non-null  float64
 13  condition_code  139582 non-null  int64  
 14  n_obs_used      139582 non-null  int64  
 15  n_del_obs_used  243 non-null     float64
 16  n_dop_obs_used  243 non-null     float64
dtypes: float64(14), int64(2), str(1)
memory usage: 20.7 MB
```

In [3]:
```
df.isnull().sum()
```

Out[3]:
```
full_name             0
H                   379
diameter              0
albedo             1123
a                     0
e                     0
i                     0
om                    0
w                     0
q                     0
ad                    0
per_y                 0
data_arc              5
condition_code        0
n_obs_used            0
n_del_obs_used    139339
n_dop_obs_used    139339
dtype: int64
```

In [4]:
```
# drop rows where target variable (diameter) has missing values
# drop columns representing excluded high-cost features (albedo)
# drop rows where selected feature subset has missing values
df_clean = df.dropna(subset=target_columns).drop(columns=dropped_columns).dropna(subset=selected_columns)
df_clean.head()
```

Out[4]:

	full_name	H	diameter	a	e	i	om	w	q	ad	per_y	data_arc	condition_code	n_obs_used	n_
0	1 Ceres (A801 AA)	3.34	939.400	2.766	0.0797	10.59	80.25	73.29	2.545	2.99	4.60	9520.0	0	1075	
1	2 Pallas (A802 FA)	4.12	513.000	2.770	0.2307	34.93	172.89	310.97	2.131	3.41	4.61	80828.0	0	9772	
2	3 Juno (A804 RA)	5.19	246.596	2.671	0.2557	12.99	169.81	247.90	1.988	3.35	4.37	80897.0	0	8343	
3	4 Vesta (A807 FA)	3.25	522.770	2.361	0.0902	7.14	103.70	151.47	2.148	2.57	3.63	25743.0	0	9451	
4	5 Astraea (A845 XA)	6.95	106.699	2.577	0.1876	5.36	141.45	359.38	2.093	3.06	4.14	65687.0	0	4395	

In [5]:
```
# select only target column and feature columns used for prediction model
df_subset = df_clean[selected_columns]
df_subset.head()
```

Out[5]:

	H	diameter	a	e	i	om	q	ad	per_y
0	3.34	939.400	2.766	0.0797	10.59	80.25	2.545	2.99	4.60
1	4.12	513.000	2.770	0.2307	34.93	172.89	2.131	3.41	4.61
2	5.19	246.596	2.671	0.2557	12.99	169.81	1.988	3.35	4.37
3	3.25	522.770	2.361	0.0902	7.14	103.70	2.148	2.57	3.63
4	6.95	106.699	2.577	0.1876	5.36	141.45	2.093	3.06	4.14

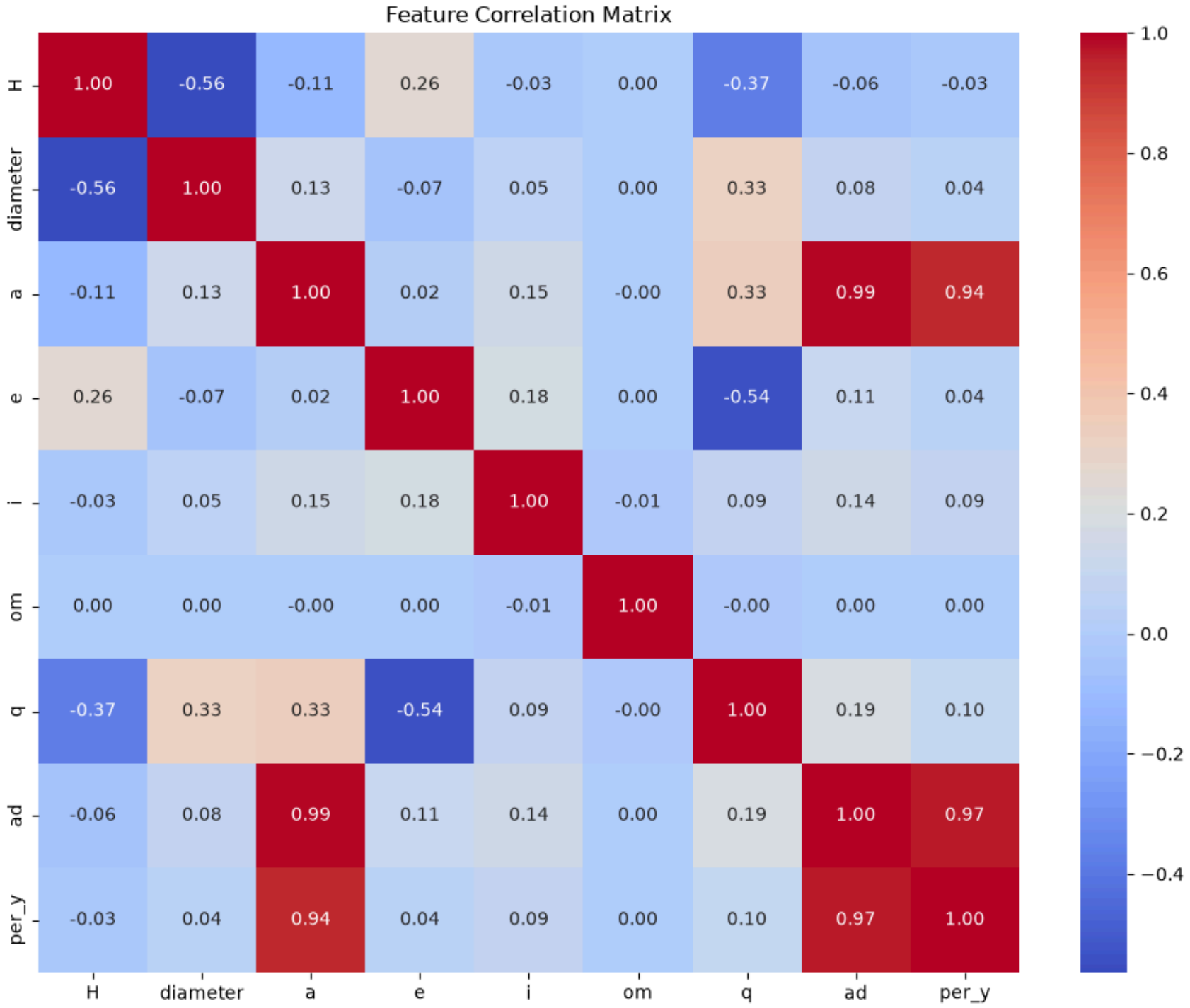
In [6]:
```
# review final subset info
df_subset.info()
```

```
<class 'pandas.DataFrame'>
Index: 139203 entries, 0 to 139581
Data columns (total 9 columns):
 #   Column    Non-Null Count   Dtype  
---  ------    --------------   -----  
 0   H         139203 non-null  float64
 1   diameter  139203 non-null  float64
 2   a         139203 non-null  float64
 3   e         139203 non-null  float64
 4   i         139203 non-null  float64
 5   om        139203 non-null  float64
 6   q         139203 non-null  float64
 7   ad        139203 non-null  float64
 8   per_y     139203 non-null  float64
dtypes: float64(9)
memory usage: 10.6 MB
```

In [7]:
```
# review final subset for null values
df_subset.isnull().sum()
```

Out[7]:
```
H           0
diameter    0
a           0
e           0
i           0
om          0
q           0
ad          0
per_y       0
dtype: int64
```

In [8]:
```
# create correlation matrix to see if any variables inherently correlate to diameter
plt.figure(figsize=(10,8))
sns.heatmap(df_subset.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Feature Correlation Matrix")
plt.tight_layout()
plt.show()
```


Feature Correlation Matrix
H
diameter
a
e
i
om
q
ad
per_y
1.00 -0.56 -0.11 0.26 -0.03 0.00 -0.37 -0.06 -0.03
-0.56 1.00 0.13 -0.07 0.05 0.00 0.33 0.08 0.04
-0.11 0.13 1.00 0.02 0.15 -0.00 0.33 0.99 0.94
0.26 -0.07 0.02 1.00 0.18 0.00 -0.54 0.11 0.04
-0.03 0.05 0.15 0.18 1.00 -0.01 0.09 0.14 0.09
0.00 0.00 -0.00 0.00 -0.01 1.00 -0.00 0.00 0.00
-0.37 0.33 0.33 -0.54 0.09 -0.00 1.00 0.19 0.10
-0.06 0.08 0.99 0.11 0.14 0.00 0.19 1.00 0.97
-0.03 0.04 0.94 0.04 0.09 0.00 0.10 0.97 1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

In [9]:
```
# save cleaned data to csv
df_subset.to_csv(dataset_cleaned_path, index=False)
```